

■ Database Availability

KEY WORDS

- Flashback Query
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 - Flashback Database
 - Flashback Drop Table
 - Cursor Sharing
-

<< Flashback Query>>

SYS@orcl> show parameter undo_retention

NAME	TYPE	VALUE
undo_retention	integer	900

SYS@orcl>

SYS@orcl> select salary from hr.emp where employee_id = 100;

SALARY
33000

SYS@orcl> select current_scn, systimestamp from v\$database;

CURRENT_SCN	SYSTIMESTAMP
5142580	24-MAY-25 12.24.06.284036 PM +09:00

SYS@orcl> update hr.emp set salary = salary * 1.1 where employee_id = 100;

1 row updated.

SYS@orcl> select salary from hr.emp where employee_id = 100;

SALARY
36300

SYS@orcl> commit;

Commit complete.

```
SYS@orcl> select salary
2  from hr.emp
3  as of timestamp to_timestamp('2025-05-24 12.24.06','yyyy-mm-dd hh24:mi:ss')
4  where employee_id = 100;
      SALARY
-----
      33000
```

```
SYS@orcl> select salary
2  from hr.emp
3  as of scn 5142580
4  where employee_id = 100 ;
      SALARY
-----
      33000
```

<< Flashback version query >>

```
SYS@orcl> drop table hr.emp purge;
```

Table dropped.

```
SYS@orcl> create table hr.emp as select * from hr.employees;
```

Table created.

```
SYS@orcl> select current_scn, checkpoint_change#, systimestamp from v$database;
```

CURRENT_SCN	CHECKPOINT_CHANGE#	SYSTIMESTAMP
5157510	5152095	24-MAY-25 03.13.36.848920 PM +09:00

```
SYS@orcl> select employee_id, last_name, salary from hr.emp where employee_id in(110,111);
```

EMPLOYEE_ID	LAST_NAME	SALARY
110	Chen	8200
111	Sciarra	7700

```
SYS@orcl> update hr.emp set salary = 1000 where employee_id = 110;
```

1 row updated.

```
SYS@orcl> delete from hr.emp where employee_id = 111;
```

1 row deleted.

```
SYS@orcl> commit;
```

Commit complete.

```
SYS@orcl> select employee_id, last_name, salary from hr.emp where employee_id in(110,111);
```

EMPLOYEE_ID	LAST_NAME	SALARY
110	Chen	1000

```
SYS@orcl> select employee_id, last_name, salary
```

```
2  from hr.emp as of timestamp to_timestamp('2025-05-24 15:13.36','yyyy-mm-dd hh24:mi:ss')
```

```
3  where employee_id in(110,111);
```

EMPLOYEE_ID	LAST_NAME	SALARY
110	Chen	8200

111 Sciarra 7700

```
SYS@orcl> select employee_id, last_name, salary
2  from hr.emp versions between timestamp
3  to_timestamp('2025-05-24 15:13.36','yyyy-mm-dd hh24:mi:ss')
4  and to_timestamp('2025-05-24 15:16:00','yyyy-mm-dd hh24:mi:ss')
5  where employee_id in(110,111);
```

EMPLOYEE_ID	LAST_NAME	SALARY
-------------	-----------	--------

111	Sciarra	7700
110	Chen	1000
110	Chen	8200
111	Sciarra	7700

```
SYS@orcl> select versions_xid, employee_id, last_name, salary
2  from hr.emp versions between timestamp
3  to_timestamp('2025-05-24 15:13.36','yyyy-mm-dd hh24:mi:ss')
4  and to_timestamp('2025-05-24 15:16:00','yyyy-mm-dd hh24:mi:ss')
5  where employee_id in(110,111);
```

VERSIONS_XID	EMPLOYEE_ID	LAST_NAME	SALARY
--------------	-------------	-----------	--------

0800180011050000	111	Sciarra	7700
0800180011050000	110	Chen	1000
	110	Chen	8200
	111	Sciarra	7700

SYS@orcl>

<< Flashback Table >>

```
SYS@orcl> select current_scn, checkpoint_change#, systimestamp from v$database;
```

```
CURRENT_SCN CHECKPOINT_CHANGE#          SYSTIMESTAMP
```

```
-----  
5158931          5152095          24-MAY-25 03.24.35.656945 PM +09:00
```

```
SYS@orcl> delete from hr.emp;
```

```
106 rows deleted.
```

```
SYS@orcl> commit;
```

```
Commit complete.
```

```
SYS@orcl> select * from hr.emp;
```

```
no rows selected
```

```
SYS@orcl> insert into hr.emp
```

```
2 select * from hr.emp as of timestamp to_timestamp('2025-05-24 15:24:30','yyyy-mm-dd hh24:mi:ss');
```

```
106 rows created.
```

```
SYS@orcl> select count(*) from hr.emp;
```

```
COUNT(*)
```

```
-----  
106
```

```
SYS@orcl> rollback;
```

```
Rollback complete.
```

```
SYS@orcl> select * from hr.emp;
```

```
no rows selected
```

```
SYS@orcl> alter table hr.emp enable row movement;
```

```
Table altered.
```

```
SYS@orcl> select row_movement from dba_tables where owner = 'HR' and table_name = 'EMP';
```

```
ROW_MOVE
```

```
-----  
ENABLED
```

```
SYS@orcl> flashback table hr.emp to timestamp  
2 to_timestamp('2025-05-24 15:24:30','yyyy-mm-dd hh24:mi:ss');  
Flashback complete.
```

```
SYS@orcl> select count(*) from hr.emp;  
COUNT(*)  
-----  
106
```

```
SYS@orcl> alter table hr.emp disable row movement;  
Table altered.
```

<< Flashback data archive >>

```
SYS@orcl> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v$datafile a,
v$tablespace b where a.ts# = b.ts#;
```

FILE#	TBS_NAME	FILE_NAME	CHECKPOINT_CHANGE#	STATUS
1	SYSTEM	/u01/app/oracle/oradata/ORCL/system01.dbf	5152095	SYSTEM
2	OETBS	/u01/app/oracle/oradata/ORCL/oetbs01.dbf	5152095	ONLINE
8	OETBS	/u01/app/oracle/oradata/ORCL/oetbs02.dbf	5152095	ONLINE
3	SYSAUX	/u01/app/oracle/oradata/ORCL/sysaux01.dbf	5152095	ONLINE
4	UNDOTBS1	/u01/app/oracle/oradata/ORCL/undotbs01.dbf	5152095	ONLINE
5	SHTBS	/u01/app/oracle/oradata/ORCL/shtbs01.dbf	5152095	ONLINE
7	USERS	/u01/app/oracle/oradata/ORCL/users01.dbf	5152095	ONLINE
9	INSA_TBS	/u01/app/oracle/oradata/ORCL/insa_tbs01	5152095	ONLINE

8 rows selected.

```
SYS@orcl> create tablespace fdatbs datafile '/u01/app/oracle/oradata/ORCL/fdatbs01.dbf' size 50m autoextend
on next 1m;
```

Tablespace created.

```
SYS@orcl> create flashback archive fda1 tablespace fdatbs quota 10m retention 1 year;
```

Flashback archive created.

```
SYS@orcl> select OWNER_NAME,FLASHBACK_ARCHIVE_NAME, RETENTION_IN_DAYS,CREATE_TIME
2 from dba_flashback_archive;
```

OWNER_NAME	FLASHBACK_ARCHIVE_NAME	RETENTION_IN_DAYS	CREATE_TIME
SYS	FDA1	1	24-MAY-25 03.35.56.000000000 PM

```
SYS@orcl> alter table hr.emp flashback archive fda1;
```

Table altered.

```
SYS@orcl> select TABLE_NAME,OWNER_NAME,FLASHBACK_ARCHIVE_NAME,ARCHIVE_TABLE_NAME,STATUS
2 from dba_flashback_archive_tables;
```

TABLE_NAME	OWNER_NAME	FLASHBACK_ARCHIVE_NAME	ARCHIVE_TABLE_NAME	STATUS
EMP	HR	FDA1	SYS_FBA_HIST_74643	ENABLED

SYS@orcl> select employee_id, salary from hr.emp where department_id = 20;

EMPLOYEE_ID SALARY

201 13000

202 6000

SYS@orcl> update hr.emp set salary = 20000 where department_id = 20;

2 rows updated.

SYS@orcl> commit;

Commit complete.

SYS@orcl> select employee_id, salary from hr.emp where department_id = 20;

EMPLOYEE_ID SALARY

201 20000

202 20000

SYS@orcl> select employee_id, salary from hr.emp as of timestamp(systimestamp-interval '5' minute) where department_id = 20;

EMPLOYEE_ID SALARY

201 13000

202 6000

SYS@orcl> ! ps -ef | grep fbda | grep -v grep

oracle 10877 1 0 15:35 ? 00:00:00 ora_fbda_orcl

SYS@orcl> alter table hr.emp no flashback archive;

Table altered.

SYS@orcl> select TABLE_NAME,OWNER_NAME,FLASHBACK_ARCHIVE_NAME,ARCHIVE_TABLE_NAME,STATUS
2 from dba_flashback_archive_tables;

TABLE_NAME OWNER_NAME FLASHBACK_ARCHIVE_NAME ARCHIVE_TABLE_NAME STATUS

EMP HR FDA1 SYS_FBA_HIST_74643 DISABLED

SYS@orcl> select * from dba_flashback_archive;

OWNER_NAME FLASHBACK_ARCHIVE_NAME FLASHBACK_ARCHIVE# RETENTION_IN_DAYS CREATE_TIME


```
-----
LAST_PURGE_TIME  STATUS
-----
```

```
SYS                FDA1                1                365        24-MAY-25 03.35.56.000000000 PM
24-MAY-25 03.35.56.000000000 PM
```

```
SYS@orcl> alter flashback archive fda1 modify retention 2 year;
Flashback archive altered.
```

```
SYS@orcl> select * from dba_flashback_archive;
```

```
OWNER_NAME  FLASHBACK_ARCHIVE_NAME  FLASHBACK_ARCHIVE#  RETENTION_IN_DAYS  CREATE_TIME
-----
```

```
LAST_PURGE_TIME  STATUS
-----
```

```
SYS                FDA1                1                730        24-MAY-25 03.35.56.000000000 PM
24-MAY-25 03.35.56.000000000 PM
```

```
SYS@orcl> select * from dba_flashback_archive_ts;
```

```
FLASHBACK_ARCHIVE_NAME  FLASHBACK_ARCHIVE#  TABLESPACE_NAME  QUOTA_IN_MB
-----
```

```
FDA1                1                FDATBS                10
```

```
SYS@orcl> alter flashback archive fda1 modify tablespace fdatbs quota 20m;
Flashback archive altered.
```

```
SYS@orcl> select * from dba_flashback_archive_ts;
```

```
FLASHBACK_ARCHIVE_NAME  FLASHBACK_ARCHIVE#  TABLESPACE_NAME  QUOTA_IN_MB
-----
```

```
SYS                FDA1                1                FDATBS                20
```

```
SYS@orcl> alter flashback archive fda1 purge before timestamp(systimestamp - interval '1' day);
Flashback archive altered.
```

```
SYS@orcl> drop flashback archive fda1;
Flashback archive dropped.
```

<< Flashback Database >>

```
SYS@orcl> select dbid, name, log_mode from v$database;
```

DBID	NAME	LOG_MODE
1727714120	ORCL	ARCHIVELOG

```
SYS@orcl> alter system set db_flashback_retention_target = 2880 scope = both;
System altered.
```

```
SYS@orcl> show parameter db_flashback_retention_target
```

NAME	TYPE	VALUE
db_flashback_retention_target	integer	2880

```
SYS@orcl> select flashback_on from v$database;
```

FLASHBACK_ON
NO

```
SYS@orcl> alter database flashback on;
Database altered.
```

```
SYS@orcl> select flashback_on from v$database;
```

FLASHBACK_ON
YES

```
SYS@orcl> show parameter db_recovery_file_dest
```

NAME	TYPE	VALUE
db_recovery_file_dest	string	/home/oracle/backup/FRA
db_recovery_file_dest_size	big integer	4G

```
SYS@orcl> select * from v$flash_recovery_area_usage;
```

FILE_TYPE	PERCENT_SPACE_USED	PERCENT_SPACE_RECLAIMABLE
NUMBER_OF_FILES	CON_ID	

CONTROL FILE	0	0	0	0
REDO LOG	0	0	0	0
ARCHIVED LOG	0	0	0	0
BACKUP PIECE	7	0	16.1	15.59
IMAGE COPY	9	0	47.37	.25
FLASHBACK LOG	2	0	.78	0
FOREIGN ARCHIVED LOG	0	0	0	0
AUXILIARY DATAFILE COPY	0	0	0	0

8 rows selected.

SYS@orcl> select current_scn, systimestamp from v\$database;

CURRENT_SCN SYSTIMESTAMP

```
-----
5167290                      24-MAY-25 04.27.30.285017 PM +09:00
```

SYS@orcl> create restore point before_hr_emp_trunc;

Restore point created.

SYS@orcl> select * from v\$restore_point;

SYS@orcl> select count(*) from hr.emp;

COUNT(*)

106

```
SYS@orcl> truncate table hr.emp;
```

Table truncated.

```
SYS@orcl> select count(*) from hr.emp;
```

```
COUNT(*)
```

```
-----
```

```
0
```

```
SYS@orcl> truncate table hr.emp;
```

Table truncated.

```
SYS@orcl> shutdown immediate
```

Database closed.

Database dismounted.

ORACLE instance shut down.

```
SYS@orcl> startup mount
```

ORACLE instance started.

Total System Global Area 1140849904 bytes

Fixed Size 8895728 bytes

Variable Size 855638016 bytes

Database Buffers 268435456 bytes

Redo Buffers 7880704 bytes

Database mounted.

```
SYS@orcl> flashback database to restore point before_hr_emp_trunc;
```

Flashback complete.

```
SYS@orcl> alter database open read only;
```

Database altered.

```
SYS@orcl> select count(*) from hr.emp;
```

```
COUNT(*)
```

```
-----
```

```
106
```

```
SYS@orcl> shutdown immediate
```

Database closed.

Database dismounted.

ORACLE instance shut down.

SYS@orcl> startup mount

ORACLE instance started.

Total System Global Area 1140849904 bytes

Fixed Size 8895728 bytes

Variable Size 855638016 bytes

Database Buffers 268435456 bytes

Redo Buffers 7880704 bytes

Database mounted.

SYS@orcl> SYS@orcl> alter database open resetlogs;

Database altered.

SYS@orcl> select count(*) from hr.emp;

COUNT(*)

106

SYS@orcl> drop restore point before_hr_emp_trunc;

Restore point dropped.

SYS@orcl> alter database flashback off;

Database altered.

SYS@orcl>

<< Flashback Drop Table >>

SYS@orcl> conn hr/hr

Connected.

HR@orcl> drop table emp;

Table dropped.

HR@orcl> show recyclebin

ORIGINAL NAME	RECYCLEBIN NAME	OBJECT TYPE	DROP TIME

EMP	BIN\$Nd53TdUGRT/gY4aJEKyAzg==\$0	TABLE	2025-05-24:17:26:05

HR@orcl> SELECT

object_name,	-- Recycle Bin name of the object
original_name,	-- Original name of the object
operation,	-- Operation type (e.g., DROP)
type,	-- Object type (e.g., TABLE, INDEX)
ts_name,	-- Tablespace name
createtime,	-- Creation time
droptime	-- Drop time

FROM

recyclebin;

OBJECT_NAME	ORIGINAL_NAME	OPERATION	TYPE	TS_NAME	CREATETIME	DROPTIME

BIN\$Nd53TdUGRT/gY4aJEKyAzg==\$0	EMP	DROP	TABLE	SYSAUX	2025-05-24:15:13:24	2025-05-24:17:26:05

HR@orcl> FLASHBACK TABLE hr.emp TO BEFORE DROP;

Flashback complete.

- FLASHBACK TABLE " BIN\$Nd53TdUGRT/gY4aJEKyAzg==\$0 " TO BEFORE DROP RENAME TO emp;
- FLASHBACK TABLE emp TO BEFORE DROP RENAME TO emp1;
- FLASHBACK TABLE emp, test_backup TO BEFORE DROP;

HR@orcl> Select count(*) from emp;

COUNT(*)

106

<< Cursor_sharing >>

```
SYS@orcl> SHOW PARAMETER cursor_sharing;
```

NAME	TYPE	VALUE
------	------	-------

cursor_sharing	string	EXACT
----------------	--------	-------

```
SYS@orcl> select * from hr.employees
```

```
2 where last_name='Lee';
```

EMPLOYEE_ID	FIRST_NAME	LAST_NAME
-------------	------------	-----------

EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID
-------	--------------	-----------	--------

SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
--------	----------------	------------	---------------

165	David	Lee	
DLEE	011.44.1346.529268	23-FEB-08	SA_REP
6800	.1	147	80

```
SYS@orcl> select * from hr.employees
```

```
2 where last_name='Lorentz';
```

EMPLOYEE_ID	FIRST_NAME	LAST_NAME
-------------	------------	-----------

EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID
-------	--------------	-----------	--------

SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
--------	----------------	------------	---------------

107	Diana	Lorentz	
DLORENTZ	590.423.5567	07-FEB-07	IT_PROG
4200		103	60

```
SYS@orcl> SELECT sql_text, executions, parsing_schema_name
```

```
2 FROM v$sql
```

```
3 WHERE sql_text LIKE 'select * from hr.employees where last_name=%';
```

SQL_TEXT	EXECUTIONS	PARSING_SCHEMA_NAME
----------	------------	---------------------

select * from hr.employees where last_name='Lorentz'	1	
--	---	--

```
SYS
```

```
select * from hr.employees where last_name='Lee'    1
SYS
```

```
SYS@orcl> ALTER SYSTEM SET cursor_sharing = FORCE ;
System altered.
```

```
SYS@orcl> select * from hr.employees
```

```
2  where last_name='Bell';
```

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
192	Sarah	Bell	SBELL	650.501.1876	04-FEB-04	SH_CLERK	4000			
				123	50					

```
SYS@orcl> select * from hr.employees
```

```
2  where last_name='Matos';
```

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
143	Randall	Matos	RMATOS	650.121.2874	15-MAR-06	ST_CLERK	2600			
				124	50					

```
SYS@orcl> SELECT sql_text, executions, parsing_schema_name
```

```
2  FROM v$sql
```

```
3  WHERE sql_text LIKE 'select * from hr.employees where last_name=%';
```

SQL_TEXT	EXECUTIONS	PARSING_SCHEMA_NAME
----------	------------	---------------------

```
select * from hr.employees where last_name='Lorentz'    1
SYS
```



```
select * from hr.employees where last_name='Lee'      1  
SYS
```

```
select * from hr.employees where last_name=:SYS_B_0  2  
SYS
```